

Chapter Two

Ethical and Social Issues In Information Systems

Ethics

- It is Principles of right and wrong that individuals, acting as free moral agents, use to make choices to guide their behavior.
- Information systems and ethics:- Information systems raise new ethical questions because they create opportunities for:
 - ❖ Intense social change, threatening existing distributions of power, money, rights, and obligations.
 - ❖ New kinds of crime.

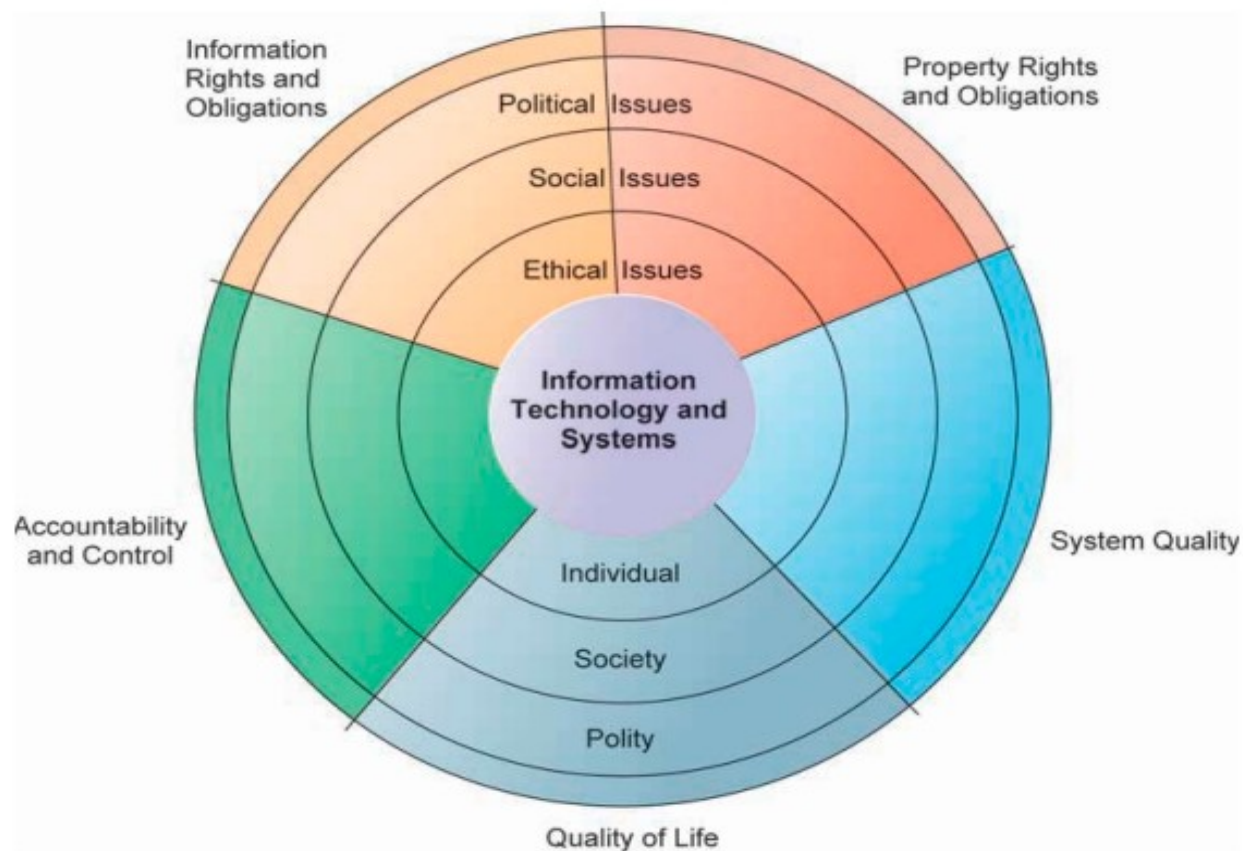
Moral dimensions of the information age

- Mainly there are **five** moral dimensions of the information age.
- The **introduction** of new information technology has a ripple **effect, raising new ethical, social, and legal issues** that must be dealt with on the individual, social, and political levels.

Moral dimensions of the information age...

- These issues have **five** moral dimensions:
 - ✓ information rights and obligations,
 - ✓ property rights and obligations,
 - ✓ system quality, quality of life, and
 - ✓ accountability and control.

Moral dimensions of the information age...



Moral dimensions of the information age...

1. Information rights and obligations:-
What information rights do individuals and organizations have with respect to themselves? What can they protect?

e.g. an employer using software to track every key stroke or website visit of a remote employee without their consent. Does the employee have a right to privacy during work hours?

Moral dimensions of the information age...

2. Property rights and obligations:- How will traditional intellectual property rights be protected in a digital society in which tracing and accounting for ownership are difficult and ignoring such property rights is so easy?

e.g. a student “cracked ” a version of expensive software instead of paying for a license. How does this impact the developer’s ability to create future innovations?

Moral dimensions of the information age...

3. Accountability and control:- Who can and will be held accountable and liable (responsible) for the harm done to individual and collective information and property rights?

e.g. If AI driven medical diagnosis tool gives a wrong prescription, who is liable? The doctor who use the program? The programmer who wrote the code? Or the hospital that bought it?

Moral dimensions of the information age...

4. System quality:- What standards of data and system quality should we demand to protect individual rights and the safety of society?

e.g. a bug in a bank's database deletes customer transaction records. What is the "acceptable" level of error before a system is considered unethical to use?

Moral dimensions of the information age...

5. Quality of life: What values should be preserved in an information- and knowledge-based society?

- ✓ Which institutions should we protect from violation?

- ✓ Which cultural values and practices are

Moral dimensions of the information age...

e.g. the “always on” culture where employees feel obligated to answer work telegram messages at 9:00 pm because technology allows them to be reached anywhere.

Nb. Technology act like a stone thrown into a pond, creating ripples that disturb existing social and political arrangements.

Key technology trends that raise ethical issues

- **Computing power doubles every 18 months:** leading to a massive organizational dependency on computer systems for critical operations.
- **Data storage costs rapidly declining:** Organizations can easily maintain detailed databases on individuals.

Key technology trends that raise ethical issues..

- **Data analysis advances**

- ✓ Greater ability to find detailed personal information on individuals
- ✓ Profiling and nonobvious relationship awareness (NORA).

NORA: Combining data from **multiple sources** to find obscure (ambiguous) hidden connections that might help identify **criminals or terrorists.**

Key technology trends that raise ethical issues

- **Networking advances and the Internet:** Copying data from one location to another and accessing personal data from remote locations are much easier.

Basic concepts for ethical analysis

- Basic concepts for ethical analysis (key elements in ethical action):
 - ✓ **Responsibility:** Accepting the potential costs, duties, and obligations for decisions you make.
 - ✓ **Accountability:** Mechanisms for identifying **responsible parties**.

Basic concepts for ethical analysis...

- ✓ **Liability:** Permits individuals (and firms) to **recover damages** done to them by other actors, systems, or organizations.
- ✓ **Due process:** a process in which laws are known and understood, and there is an ability to **appeal (request) to higher authorities** to ensure that the laws are applied correctly.
- ✓ **Individuals** can **recover damages** done to them through a set of laws characterized by due process.

Steps of ethical analysis

- Ethical analysis have 5 steps:
 1. Identify and clearly describe the facts
 2. Define the conflict or dilemma and identify the higher-order values involved.
 3. Identify the stakeholders
 4. Identify the options that you can reasonably take
 5. Identify the potential consequences of your options.

Steps of ethical analysis...

1. Identify and clearly describe the facts

- Find out **who did what to whom, and where, when, and how.**
- It also helps to get the **opposing parties** involved in an ethical dilemma or conflict to agree on the facts.

Steps of ethical analysis...

2. Define the conflict or dilemma and identify the higher-order values involved.

- Ethical, social, and political issues always reference **higher values**.
- **example**:-freedom, privacy, protection of property, and the free enterprise system.

Steps of ethical analysis...

3. Identify the stakeholders: Every ethical, social, and political issue has stakeholders:

- players in the game who have an interest in the outcome, who have invested in the situation, and usually who have vocal opinions.
- Find out the identity of these groups and what they want.
- This will be useful later when designing a solution.

Steps of ethical analysis...

4. Identify the options that you can reasonably take

- You may find that **none of the options satisfy all** the interests involved, but that **some options** do a **better job than others**.
- Sometimes arriving at a good or ethical solution may not always be a balancing of consequences to stakeholders.

Steps of ethical analysis...

5. Identify the potential consequences of your options.

- Some options may be ethically correct but disastrous from other points of view.
- Other options may work in one instance but not in other similar instances.
- Always ask yourself, “What if I choose this option consistently over time?”

Internet Challenges to Privacy

Privacy

- Claim of individuals to be **left alone**, free from **surveillance** or **interference** from **other individuals**, organizations, or the state.
- Ability to control **information about yourself**.

E.g. let you will search books that assist me on exam on google after a while there are many options comes to in your website based on your search history with out your consent...this violates the rights of free from surveillance.

Internet Challenges to Privacy...

- **Cookies**
 - ✓ Tiny files downloaded by Web site to visitor's hard drive
 - ✓ Identify visitor's browser and track visits to site
 - ✓ Allow Web sites to develop profiles on visitors

e.g. if you ordered online goods from Amazon earlier the cookies save the data for the next day if you will visit the page the website says welcome then it will list the choices of you that are related with your previous choice so this implies the cookies track your status quo on the website.

Internet Challenges to Privacy...

- **Web bugs** (Web beacons): tiny graphics embedded in **e-mail** messages and **Web pages**.
 - ✓ Designed to **monitor who is reading** a message and **transmitting** that information to another computer on the Internet.

e.g. if a super market sends an email to its customer by saying “big sale” then when the customer opens the email the web bug is already loaded on the supermarket’s server so they track who, when, where does open the email by computer or phone so the customer will track by the market with out his knowledge.

Internet Challenges to Privacy...

- **Spyware:** Secretly installed on user's computer
 - ✓ May transmit user's **keystrokes** or display **unwanted ads**.
- E.g. if you can download a free film then hackers load spyware on your computer without your knowledge they will save the password that you insert for your accounts including bank account then it transfers or sends the recorded data for the previous hackers.

Property Rights

- **Intellectual property:** **Intangible property** of any kind created by individuals or corporations.
- **Three** ways that intellectual property is protected
 - ✓ **Trade secret:** Intellectual work or product belonging to business, **not** in the **public domain**.
 - ✓ Software that contains novel or unique elements, procedures, or compilations can be included as a trade secret.
 - ✓ E.g. coca cola formula

Property Rights...

✓ **Copyright:** Statutory **grant** protecting intellectual property from **being copied** for the **life of the author**, plus 70 years.

e.g. books author, poem and lyrics writer
right

✓ **Patents:** **Grants creator of invention** an exclusive monopoly on ideas behind invention for 20 years.

e.g. innovations related with engineering
inventions

Challenges to Intellectual Property Rights

- Digital media different from physical media (e.g. books)
 - Ease of replication
 - Ease of transmission (networks, Internet)
 - Difficulty in classifying software
 - Compactness
 - Difficulties in establishing uniqueness

Principal sources of poor system performance

- Three principal sources of poor system performance.:
 - ✓ Software bugs, errors
 - ✓ Hardware or facility failures
 - ✓ Poor input data quality (most common source of business system failure)

Negative social consequences of systems

- **Computer crime and abuse**

✓ **Computer crime:** **illegal act** through use of computer or against a computer system.

✓ computer may be **object** or **instrument** of crime.

e.g hacking

Negative social consequences of systems...

- ✓ **Computer abuse:** unethical acts, **not illegal**
e.g. an organizational employee scroll social media by using organizations computer during work time.
- ✓ **Spam:** high costs for businesses in dealing with spam
- ✓ Spam is **junk e-mail sent** by an **organization or individual to a mass audience of Internet users** who have expressed **no interest** in the product or service being marketed.

Negative social consequences of systems...

- **Dependence and vulnerability:** public and private organizations ever **more dependent** on computer systems, therefore, they are highly **vulnerable** if these **systems fail**.
- **Maintaining boundaries:** Technology impact the **balance** between Family, Work, and Leisure.
- Equity and access: the digital divide.

End of chapter 2
thank you!